

TASK-1:- VULNERABILITY ASSESMENT REPORT

CIN:- FIT/APR26/CS7953

EXECUTIVE SUMMARY

This report documents a vulnerability assessment conducted on the live web application. The assessment was performed using industry-standard security tools including Nmap, OWASP ZAP, cURL, Browser Developer Tools, and manual inspection techniques. The objective of this assessment was to identify security misconfigurations, exposed services, missing security headers, and other web security issues that could potentially affect confidentiality, integrity, and availability.

The assessment was conducted using passive and non-intrusive testing methods only. No exploitation or destructive actions were performed during the assessment.

ASSESSMENT SCOPE

- Target URL: <https://lets-chat-delta.vercel.app>
- Assessment Type: Passive Vulnerability Assessment
- Tools Used: Nmap, OWASP ZAP, cURL, Chrome DevTools
- Testing Approach: Reconnaissance, Passive Scanning, Manual Inspection

NMAP SCAN RESULTS

The Nmap scan identified the following exposed ports and services:

Port	State	Service
80/tcp	Open	HTTP
443/tcp	Open	HTTPS

ASSESSMENT SCREENSHOTS:-

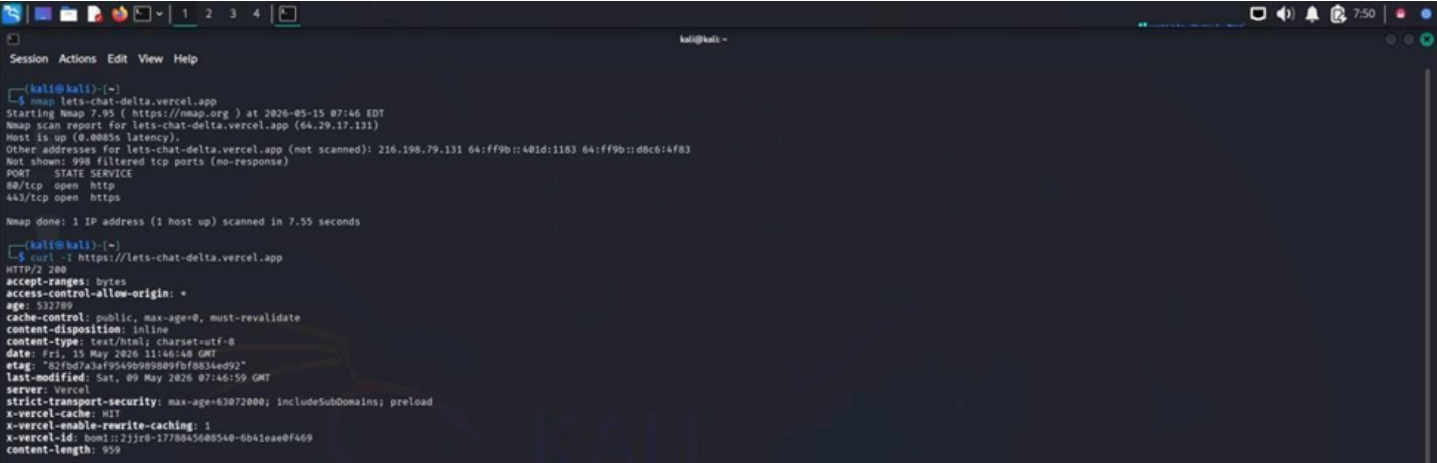


Figure 1: Nmap Scan Output

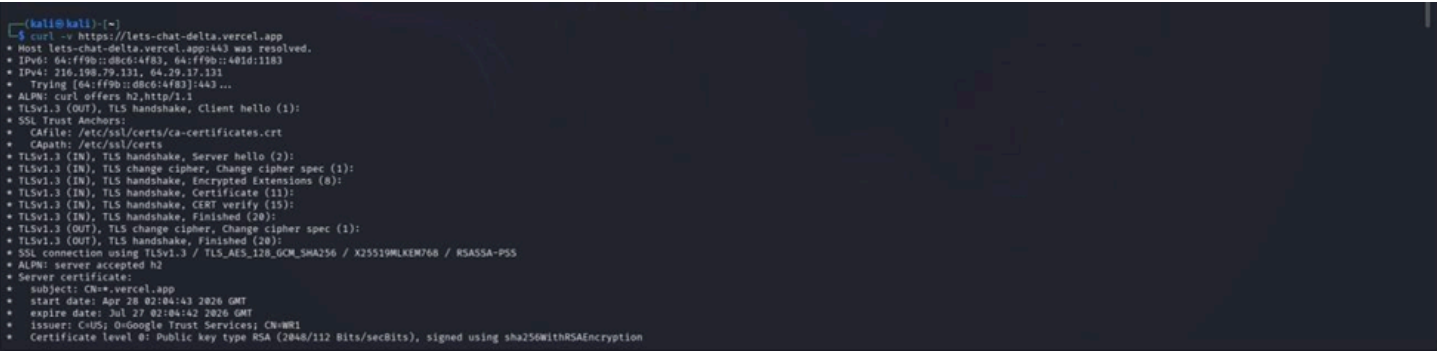


Figure 2: cURL SSL/TLS Verification

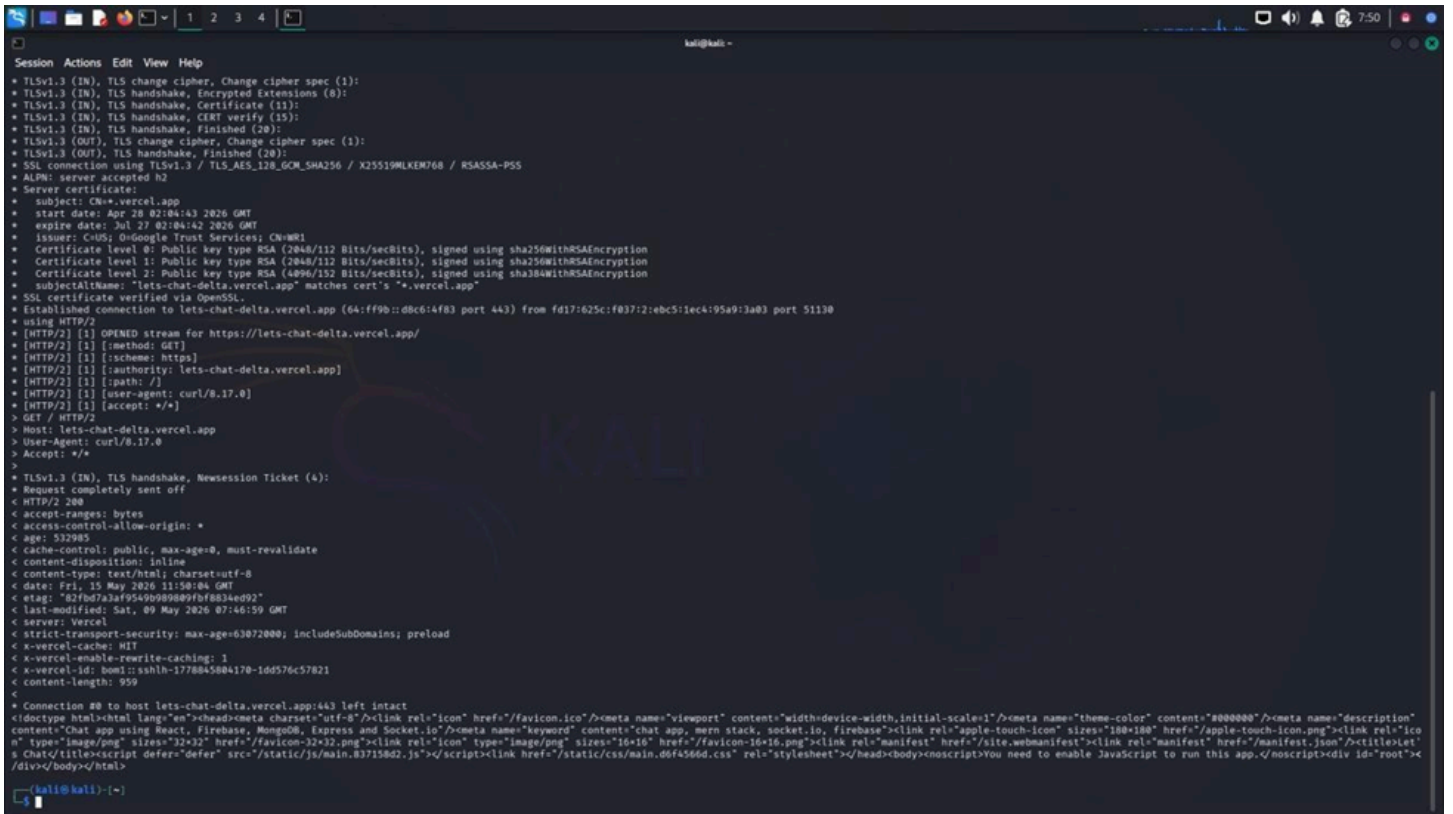


Figure 3: Browser Developer Tools Network Analysis

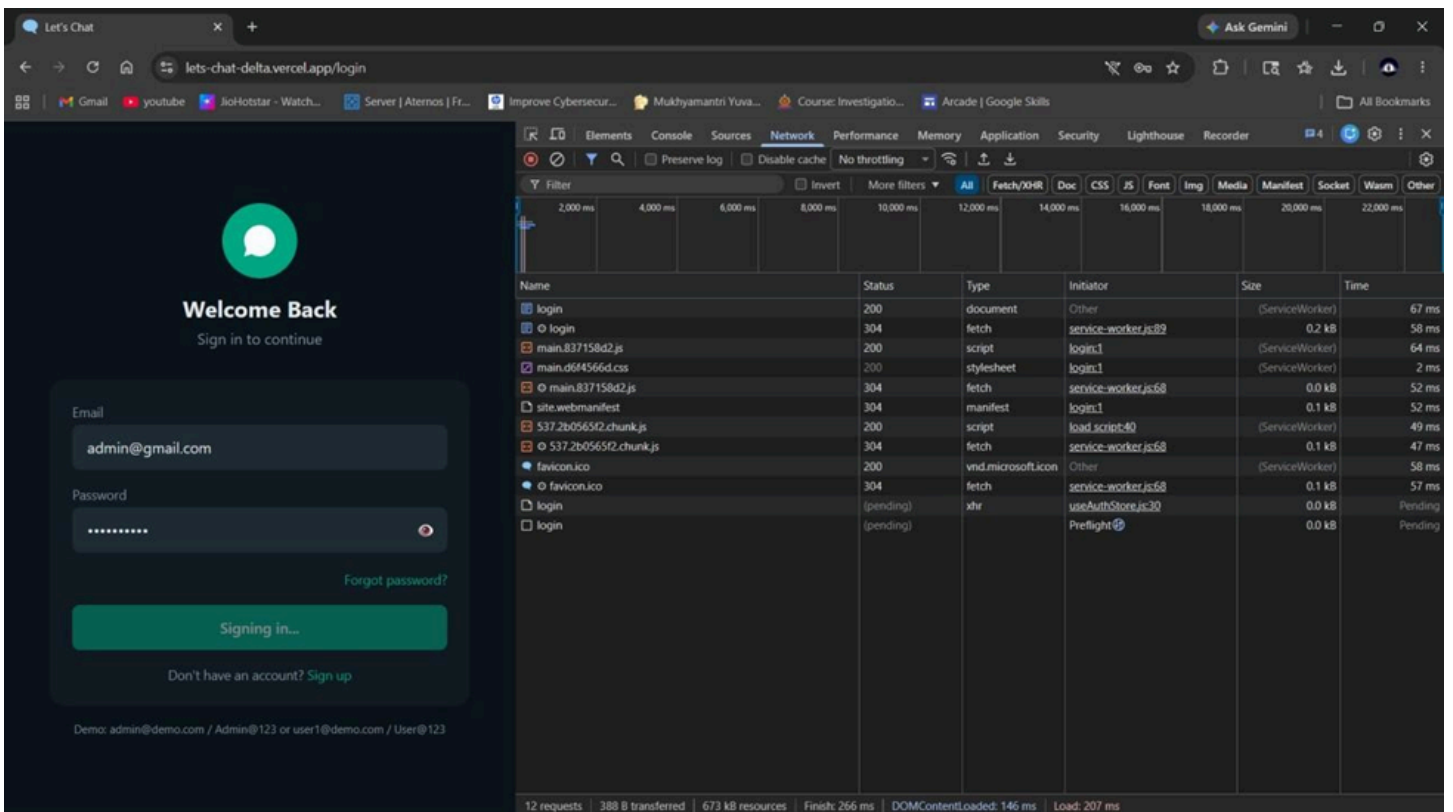


Figure 4: OWASP ZAP Passive Scan Findings

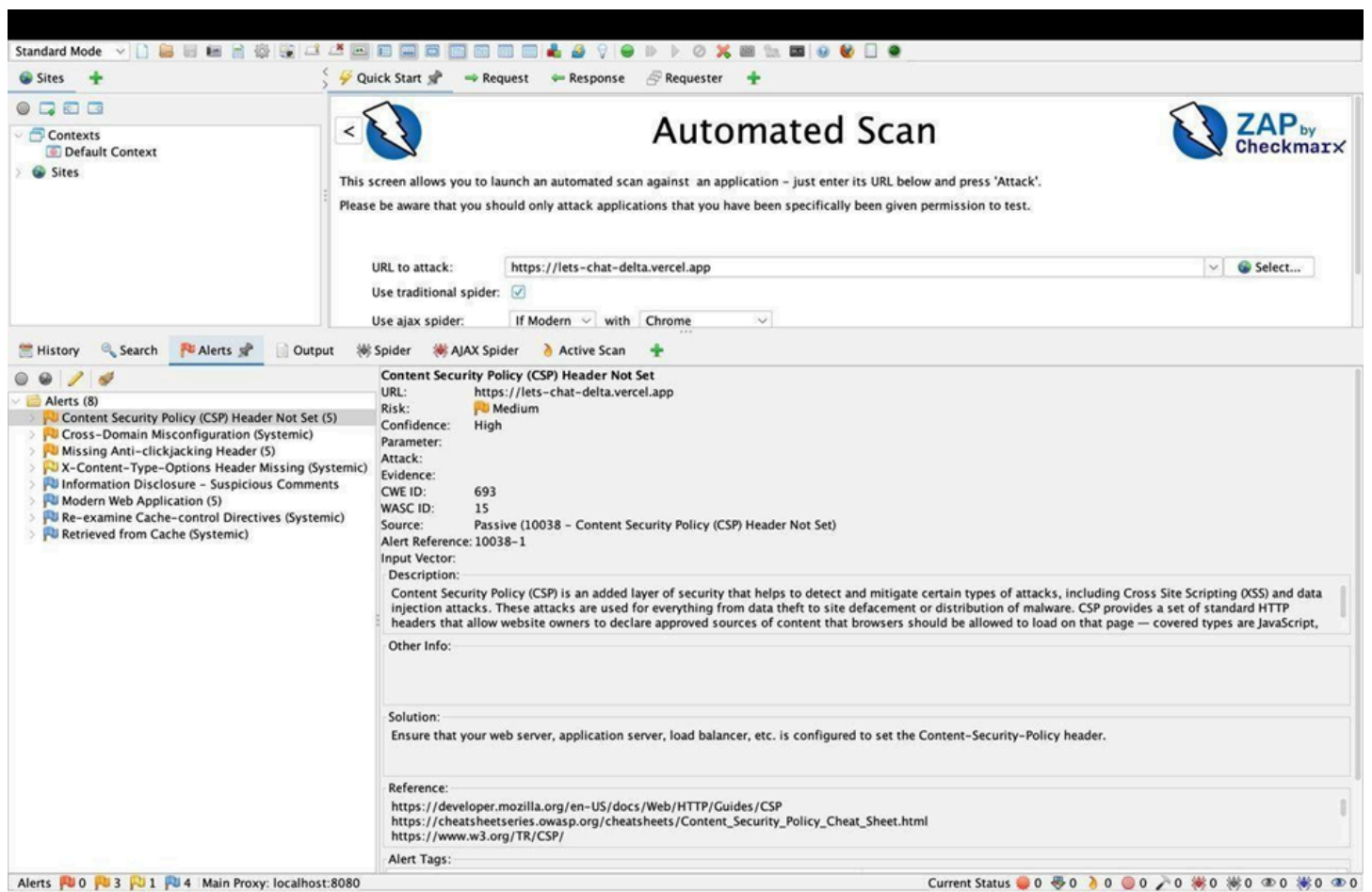


Figure 5: Login Interface of Target Application

RECOMMENDATIONS

- Implement a strong Content Security Policy (CSP).
- Add all recommended HTTP security headers.
- Enable secure cookie flags including HttpOnly and Secure.
- Regularly perform vulnerability assessments and penetration testing.
- Monitor logs and suspicious activities continuously.
- Apply security hardening and dependency updates regularly.

CONCLUSION

The assessment identified several medium and low-risk vulnerabilities related to security headers and application configuration. While no critical vulnerabilities were exploited during

this assessment, implementing the recommended mitigations will significantly improve the overall security posture of the application. Regular assessments, patch management, and secure development practices are strongly recommended.